



Exploring Skewness and Outlier Detection in Healthcare Data

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INTRODUCTION

Healthcare measurements such as BMI, glucose and blood pressure often show skewed distributions with extreme values. Undetected outliers distort averages and weaken the reliability of downstream analysis.

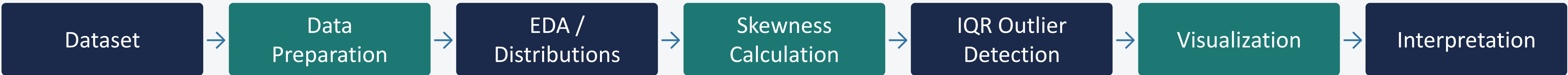
DATASET

- Synthetic healthcare dataset
- n = 200 records
- Variables: Age, BMI, Glucose, Systolic BP
- Sample data - not real patient records

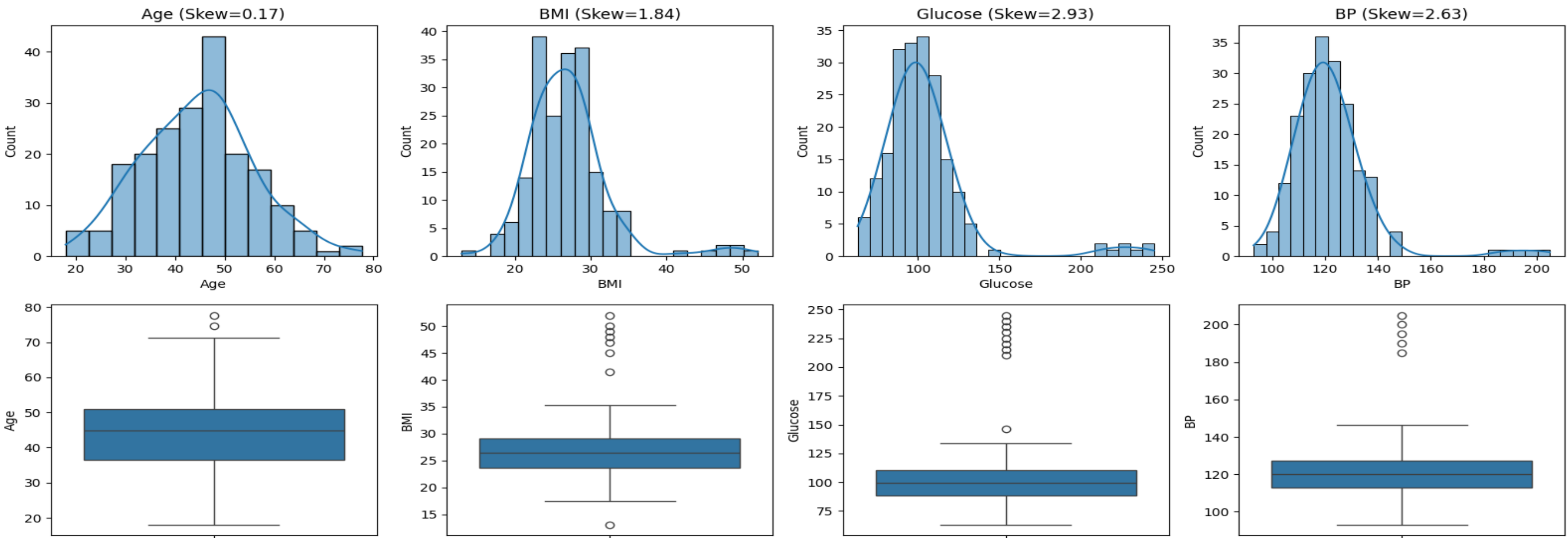
OBJECTIVES

- Analyze distribution shape using skewness
- Detect outliers using the IQR method
- Interpret their effect on healthcare-data analysis

METHODOLOGY / PIPELINE



RESULTS



	Age	BMI	Glucose	BP
0	50.960570	27.431149	117.691602	115.406391
1	43.340828	28.243138	92.962365	111.501556
2	52.772262	30.332205	74.302982	128.303358
3	63.276358	30.215208	120.308086	111.439162
4	42.190160	20.489323	98.281902	120.715662

Variable	Skewness	Interpretation	Outliers
Age	0.17	Approximately symmetric	2
BMI	1.84	Right-skewed	8
Glucose	2.93	Highly right-skewed	9
BP	2.63	Highly right-skewed	5

KEY RESULT

Glucose showed the highest positive skewness (2.93) and the highest number of IQR-detected outliers (9).

Python Implementation:

```
import pandas as pd, numpy as np
np.random.seed(42)
n = 200

df = pd.DataFrame({
    "Age": np.random.normal(45,12,n).clip(18,90),
    "BMI": np.r_[np.random.normal(26,4,n-6),48,50,52,45,47,49],
    "Glucose": np.r_[np.random.normal(100,15,n-8),220,230,240,210,225,235,215,245],
    "BP": np.r_[np.random.normal(120,10,n-5),190,195,185,200,205]
})

print(df.skew().round(2))

for c in df:
    q1,q3 = df[c].quantile([.25,.75])
    iqr = q3-q1
    out = df[(df[c]<q1-1.5*iqr) | (df[c]>q3+1.5*iqr)]
    print(c, len(out))
```

KEY FINDINGS

- Age was approximately symmetric
- BMI, Glucose and BP were positively skewed
- Glucose showed the strongest skewness
- IQR identified statistically unusual observations

CONCLUSION

Skewness and IQR-based outlier detection reveal that clinical variables like glucose and BP deviate strongly from normality. Identifying these extremes before analysis is essential to avoid biased summary statistics in healthcare data.

FUTURE SCOPE

- Larger clinical datasets
- Automated anomaly detection
- Predictive healthcare analytics